2.4.15:

Solution:

Proof. ($\Rightarrow$) Suppose that T is an isomorphism. Since T is injective, $T(\beta)$ contains exactly n vectors, so it is sufficient to prove that $T(\beta)$ is linearly independent. Write $\beta = \{b_1, \ldots, b_n\}$ and suppose $\sum_{i=1}^{n} a_i T(b_i) = \vec{0}$. By linearity, $T\left(\sum_{i=1}^{n} a_i b_i\right) = \vec{0}$, so by injectivity $\sum_{i=1}^{n} a_i b_i = \vec{0}$. Since β is linearly independent, every $a_i = 0$. Thus, $T(\beta)$ is linearly independent and thus a basis for W. ($\Leftarrow$) Suppose $T(\beta)$ is a basis for W. It is sufficient to show that T is a bijection to prove that it is an isomorphism, as if T is bijective, then it is invertible. It is clear that T is onto: since $T(\beta)$ spans W, every $w \in W$ can be written $w = \sum a_i T(b_i) = T\left(\sum a_i b_i\right)$ and so lies in the image of T. Now, since $\dim V = \dim W$, by Thm 2.5, T is also one-to-one. Thus T is an isomorphism.

□

2.5.2:

Solution:

(b) We have that

$$\begin{pmatrix} 0 \\ 10 \end{pmatrix} = 4\begin{pmatrix} -1 \\ 3 \end{pmatrix} + 2\begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 5 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} + 3\begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

so,

$$\boxed{Q = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}}$$

(c) We have that

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 3\begin{pmatrix} 2 \\ 5 \end{pmatrix} + 5\begin{pmatrix} -1 \\ -3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 2 \\ 5 \end{pmatrix} - 2\begin{pmatrix} -1 \\ -3 \end{pmatrix}$$

so

$$\boxed{Q = \begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix}}$$

2.5.3*c*:

Solution: Since $P_2(\mathbb{R}) \cong \mathbb{R}^3$ we know

$$\beta = \left\{ \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \beta' = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

and,

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = -\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + 0\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

so,

$$\boxed{Q = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ -3 & 2 & 1 \end{pmatrix}}$$

2.5.5:

Solution: Directly applying the theorem we see

$$[T]_{\beta'} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \boxed{\begin{pmatrix} 1/2 & -1/2 \\ 1/2 & -1/2 \end{pmatrix}}$$

2.5.6:

Solution:

(b) We first find Q by letting Q be the 2×2 matrix whose jth column is the jth vector of β:

$$\boxed{Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}$$

Then, by the corollary to Thm 2.23:

$$[L_A]_\beta = Q^{-1}AQ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \boxed{\begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}}$$

(c) As above we have

$$\boxed{Q = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}}$$

so,

$$\begin{aligned}(Q \mid I_3) = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) &\xrightarrow{r_2 \leftrightarrow r_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \\ &\xrightarrow{r_3 \to r_3 - r_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \\ &\xrightarrow{r_2 \to r_2 - r_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \\ &\xrightarrow{r_1 \to r_1 - r_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) = (I_3 \mid Q^{-1})\end{aligned}$$

so

$$[L_A]_\beta = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix} = \boxed{\begin{pmatrix} 2 & 2 & 2 \\ -2 & -3 & -4 \\ 1 & 1 & 2 \end{pmatrix}}$$

3.2.5*a*:

Solution: We have

$$\begin{pmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix} \xrightarrow[r_3 \to r_3 - r_1]{r_2 \to r_2 + r_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow[r_1 \to r_1 - 2r_2]{r_2 \to \frac{1}{3} r_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow[r_2 \to -\frac{1}{2} r_2]{r_2 \leftrightarrow r_3} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$
$$\xrightarrow{r_3 \to r_3 - r_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$\xrightarrow{r_1 \to r_1 + r_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so, rank $(()\, A) = 3$. Thus,

$$(A \mid I_3) = \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -1 & 1 & 2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/6 & -1/3 & 1/2 \\ 0 & 1 & 0 & 1/2 & 0 & -1/2 \\ 0 & 0 & 1 & -1/6 & 1/3 & 1/2 \end{array}\right) = (I_3 \mid A^{-1})$$

by the same sequence of elementary row operations as above. Hence,

$$\boxed{A^{-1} = \begin{pmatrix} 1/6 & -1/3 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/6 & 1/3 & 1/2 \end{pmatrix}}$$

3.2.6*a*:

Solution: Let β be the standard ordered basis of $P_2(\mathbb{R})$ so

$$[T]_\beta = \begin{pmatrix} -1 & 2 & 2 \\ 0 & -1 & 4 \\ 0 & 0 & -1 \end{pmatrix}$$

since $\det([T]_\beta) \neq 0$, it follows that $([T]_\beta)^{-1}$ exists.

$$([T]_\beta \mid I_3) = \left(\begin{array}{ccc|ccc} -1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -1 & 4 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow[r_3 \to -r_3]{r_1 \to -r_1,\ r_2 \to -r_2} \left(\begin{array}{ccc|ccc} 1 & -2 & -2 & -1 & 0 & 0 \\ 0 & 1 & -4 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right)$$

$$\xrightarrow[r_1 \to r_1 + 2r_3]{r_2 \to r_2 + 4r_3} \left(\begin{array}{ccc|ccc} 1 & -2 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -1 & -4 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right)$$

$$\xrightarrow{r_1 \to r_1 + 2r_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -2 & -10 \\ 0 & 1 & 0 & 0 & -1 & -4 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right) = (I_3 \mid ([T]_\beta)^{-1})$$

so,

$$\boxed{([T]_\beta)^{-1} = \begin{pmatrix} -1 & -2 & -10 \\ 0 & -1 & -4 \\ 0 & 0 & -1 \end{pmatrix}}$$

and

$$\boxed{T^{-1}(a_0 + a_1 x + a_2 x^2) = (-a_0 - 2a_1 - 10a_2) + (-a_1 - 4a_2)x - a_2 x^2}$$

3.4.6:

Solution: Notice that the first, third, and fifth columns of $B = \text{rref}(A)$ are e_1, e_2, and e_3 respectively. Since $b_2 = -3e_1$ it follows that $a_2 = -3a_1$,

$$a_2 = -3 \begin{pmatrix} 1 \\ -2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ 3 \\ -9 \end{pmatrix}$$

Similarly, $b_4 = 4e_1 + 3e_2 \Rightarrow a_4 = 4a_1 + 3a_3$,

$$a_4 = 4 \begin{pmatrix} 1 \\ -2 \\ -1 \\ 3 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ 1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 2 \\ 0 \end{pmatrix}$$

Now, $b_6 = 5e_1 + 2e_2 - e_3 \Rightarrow a_6 = 5a_1 + 2a_3 - a_5 \Rightarrow a_5 = 5a_1 + 2a_3 - a_6$,

$$a_5 = 5 \begin{pmatrix} 1 \\ -2 \\ -1 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 1 \\ 2 \\ -4 \end{pmatrix} - \begin{pmatrix} 3 \\ -9 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -3 \\ 2 \end{pmatrix}$$

Hence,

$$\boxed{A = \begin{pmatrix} 1 & -3 & -1 & 1 & 0 & 3 \\ -2 & 6 & 1 & -5 & 1 & -9 \\ -1 & 3 & 2 & 2 & -3 & 2 \\ 3 & -9 & -4 & 0 & 2 & 5 \end{pmatrix}}$$